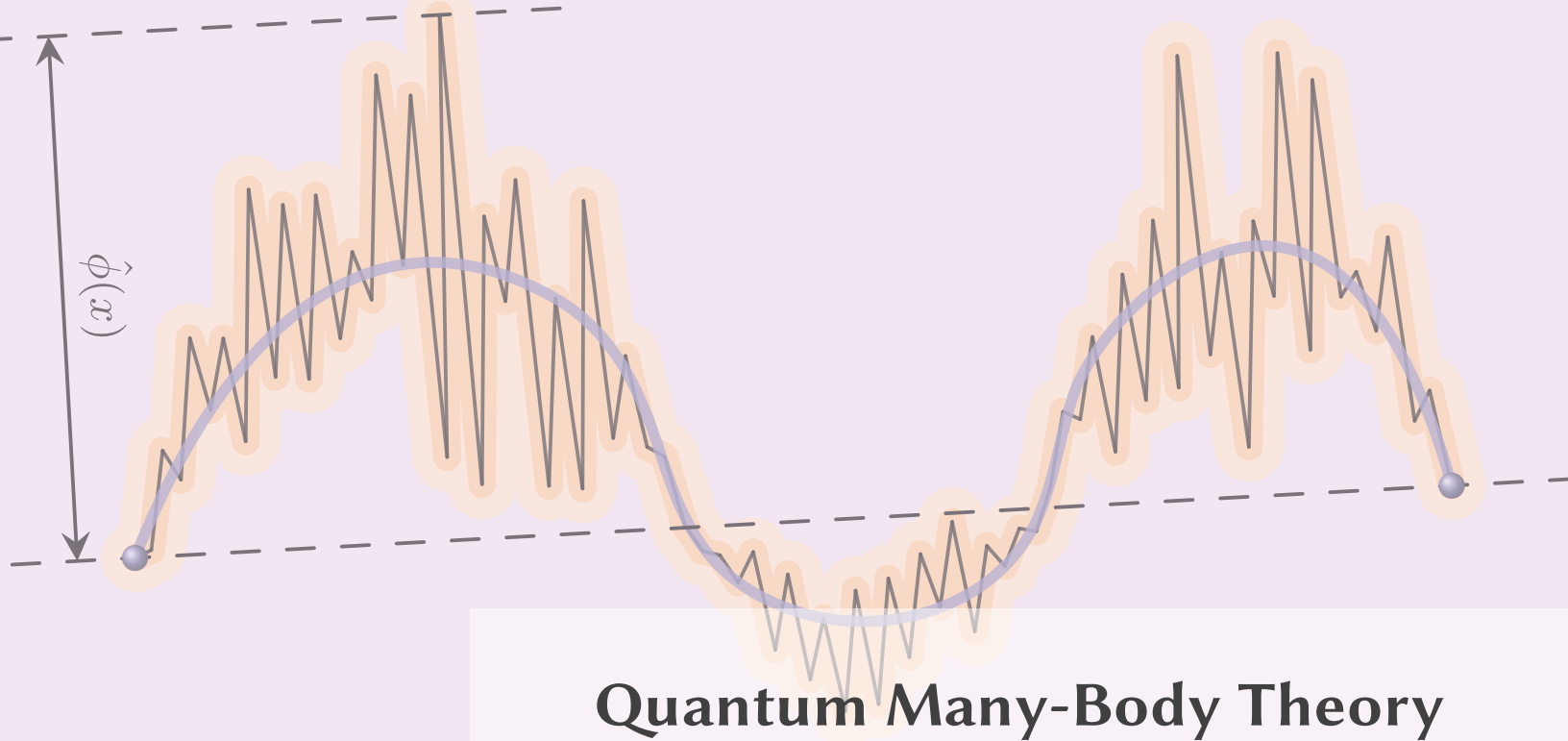




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APPENDIX A Final Project

Grading Criteria: Finishing the **Minimal** parts guarantees 85% points. With 1 step solved in **Advanced** part, you get the rest. Beyond that, you get bonuses.

Problem 0.1. Consider the following two-band/orbital Hubbard models at quarter filling on a 2D lattice.

$$\mathcal{H} = \sum_{ij,\sigma} \sum_{m=1}^2 t_{ij} c_{i,\sigma,m}^\dagger c_{j,\sigma,m} - \mu \sum_{i,\sigma} \sum_{m=1}^2 n_{i,\sigma,m} + U \sum_i \sum_{m=1}^2 n_{i,\uparrow,m} n_{i,\downarrow,m} - 2J_H \sum_i \vec{S}_{i,1} \cdot \vec{S}_{i,2} + (U' - J_H/2) \sum_i (n_{i,\uparrow,1} + n_{i,\downarrow,1})(n_{i,\uparrow,2} + n_{i,\downarrow,2}).$$

Here i, j label the lattice sites; $m = 1, 2$ is the orbital and σ the spin index. Thus, $c_{i,\sigma,m}^\dagger$ creates an electron at site i in orbital m with spin σ . As usual, μ represents the chemical potential with $n = c^\dagger c$ being the density operator.

For simplicity, it is easier to consider the cases for which there are two orbitals on each lattice site.

Brief explanation of the model: U and U' are intra- and inter-orbital Coulomb interactions; J_H is the so-called Hund's rule coupling, which is a collective result of both Coulomb repulsion and Pauli exclusion principle in the atomic states. It is a ferromagnetic spin-spin interaction between electrons on different orbitals. The last term is a constraint on the total amplitudes of U' and J_H , because they are competing against each other under the energy-minimization principle.

Relevance of this model:

- (a) It is closely related to twisted bilayer graphene at quarter filling, which is experimentally found to be a ferromagnetic anomalous Hall insulator, and is the parental state of the underlying fractional Chern insulator states. There are clues suggesting it may be a strongly correlated insulator, and this model, with proper parameters, could be the minimal model to describe this 1/4-filling state.
- (b) It could be the minimal model and parent state for multi-band 3d/4d oxides such as Sr_2RuO_4 , which is famously a ferromagnetic Fermi liquid and an interesting unconventional superconductor. Other similar compounds in the same group, such as $\text{Sr}_3\text{Ru}_2\text{O}_7$, can also be derived from this model additional tuning (327 maybe approximated by a two-layer two-band model).
- (c) It is also potentially related to other 3d unconventional superconductors with multiple active bands, including both iron pnictides, and chalcogenides and the newly discovered nickel-based family.

Remark (Expectation and hints.). At exact 1/4-filling and sufficiently large- U , it must remain a Mott insulator, but the second-order perturbation to energy will be verified to be different from 1/2-filling; both spin and orbital-pseudo-spin will be involved. Semi-classically, this is known as the Kugel-Khomskii model *K. I. Kugel and D. Khomskii, "The jahn-teller effect and magnetism: transition metalcompounds", Soviet Physics Uspekhi 25, 231 (1982).* **Note: Your results need not be exactly reproducing this results.**

Objectives:

- I. **Minimal.** Obtain the second order perturbation to energy in the large- U limit in some way. You can choose
 - (a) Conventional quantum mechanical perturbation for two sites,
 - (b) Schrieffer-Wolff transformation,
 - (c) Algebraic-dynamical method, as I demonstrated for the single-band Hubbard model
- II. **Minimal.** Discuss the semi-classical ground state configuration: Can you construct a Néel-state type product state? Make the construction or discuss why you cannot. (This is an open question already.)
- III. **Advanced.**
 - (a) Obtain the single particle Green's function on top of your semi-classical ground state. Try to make reasonable approximations if too complicated.
 - (b) Can you set up consistent equation(s) for important physical quantities such as magnetization, orbital-magnetization (or polarization), etc?
 - (c) If you reached here, congratulations, we can start to write a paper.

Solution. First, we make some preparations. Split the Hamiltonian into two terms

$$\mathcal{H} = \sum_i \mathcal{H}_i^{\text{loc}} + \mathcal{H}_t = \mathcal{H}_0 + \mathcal{V},$$

where

- (a) Atomic term $\mathcal{H}_i^{\text{loc}}$: Contains all the large energy scales (Coulomb U , U' , and Hund's J_H).
- (b) Small hopping term $\mathcal{H}_t$: Can be treated as a perturbation.

The two terms are expressed as

$$\begin{aligned} \mathcal{H}_i^{\text{loc}} &= U \sum_{m=1}^2 n_{i,\uparrow,m} n_{i,\downarrow,m} + (U' - J_H/2)(n_{i,\uparrow,1} + n_{i,\downarrow,1})(n_{i,\uparrow,2} + n_{i,\downarrow,2}) \\ &\quad - 2J_H \vec{S}_{i,1} \cdot \vec{S}_{i,2} - \mu \sum_{\sigma} \sum_{m=1}^2 n_{i,\sigma,m}, \\ \mathcal{H}_t &= \sum_{ij,\sigma} t_{ij} \sum_{m=1}^2 c_{i,\sigma,m}^{\dagger} c_{j,\sigma,m}. \end{aligned}$$

In the expression of $\mathcal{H}_t$, $c_{i,\sigma,m}^{\dagger}$ creates an electron at site i in orbital m , and $c_{j,\sigma,m}$ annihilates an electron at site j in orbital m . We label the states of the two sites as $|m_i, \sigma_i; m_j, \sigma_j\rangle$, with orbitals m_i, m_j and spins σ_i, σ_j on sites i, j . Then, one finds that only two hopping processes

$$|m, \sigma_i; m, \sigma_j\rangle \xrightarrow[\sigma_i \neq \sigma_j]{\mathcal{V}_{i \rightarrow j}} t_{ij} |0_i; m\sigma_i, m\sigma_j\rangle, \quad |m, \sigma_i; m', \sigma_j\rangle \xrightarrow{\mathcal{V}_{i \rightarrow j}} t_{ij} |0_i; m\sigma_i, m'\sigma_j\rangle$$

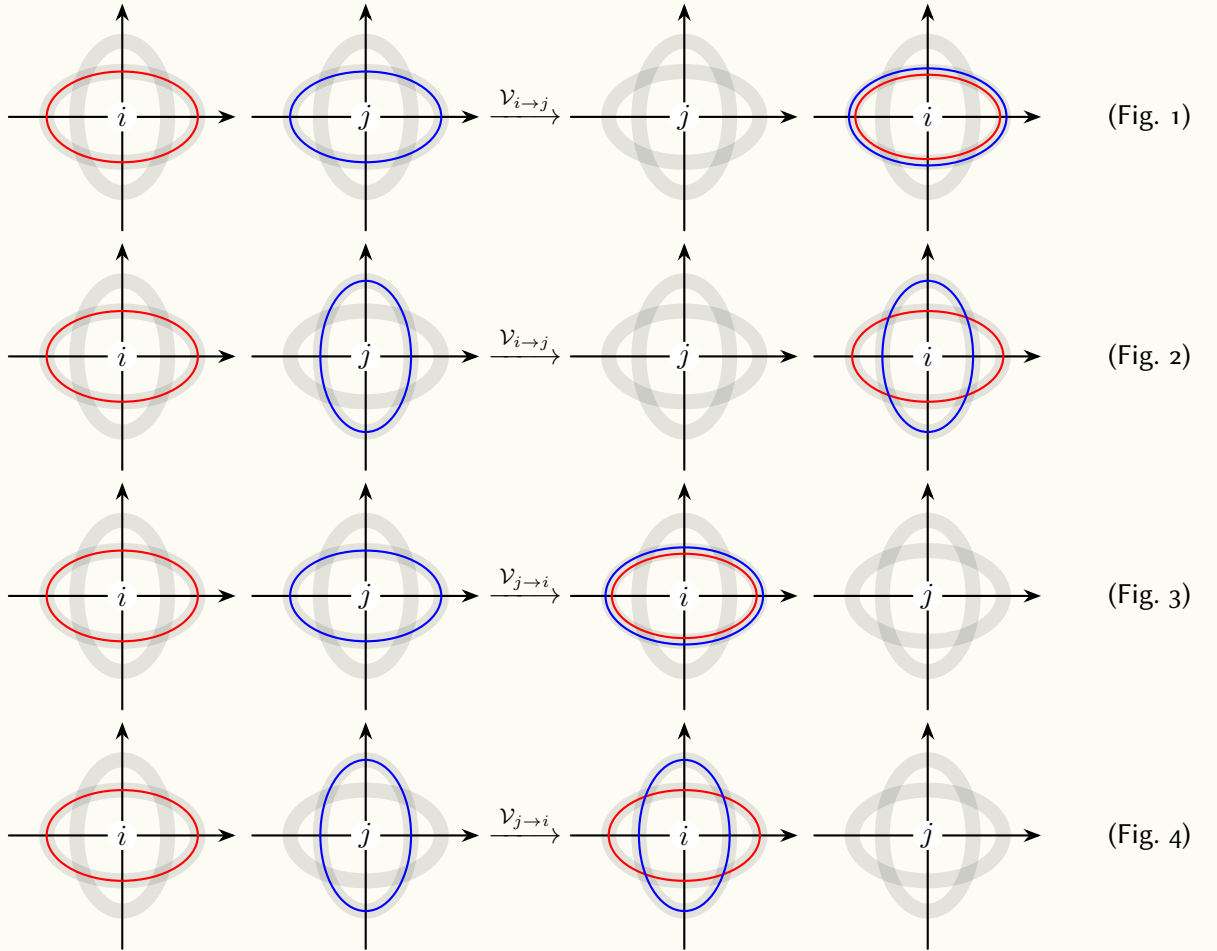
with their inversions

$$|m, \sigma_i; m, \sigma_j\rangle \xrightarrow[\sigma_i \neq \sigma_j]{V_{j \rightarrow i}} |m\sigma_i, m\sigma_j; 0_j\rangle, \quad |m, \sigma_i; m', \sigma_j\rangle \xrightarrow{V_{j \rightarrow i}} t_{ij} |m\sigma_i, m'\sigma_j; 0_j\rangle$$

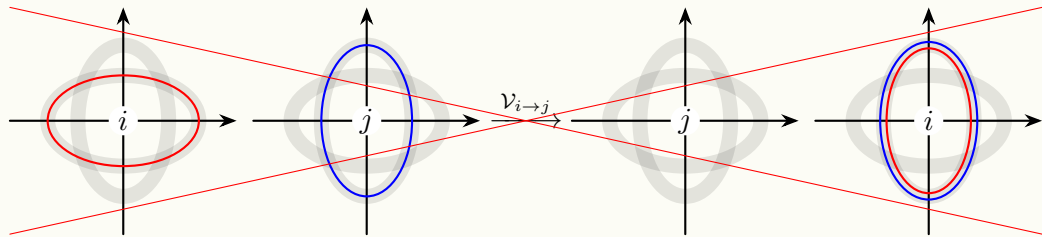
are allowed in this model, and a hopping process of the form

$$|m, \sigma_i; m', \sigma_j\rangle \xrightarrow{V_{i \rightarrow j}} t_{ij} |0_i; m'\sigma_i, m'\sigma_j\rangle$$

is not allowed. The allowed hops can also be expressed in figures



where different colors correspond to different spin directions, and the hop like



is not allowed. In the large- U limit ($t \ll U, U', J_H$), the atomic term determines the atomic eigenstates and energies. For a fixed site i , we can obtain its energy by substituting 2 different situations into $\mathcal{H}^{\text{loc}}$.

(a) **Both electrons in the same orbital m .** Due to Pauli exclusion, the spins must be opposite: One $\uparrow$ and one $\downarrow$. The four terms are

- i. $n_{\uparrow,m}n_{\downarrow,m}$ satisfies the same orbital m and different spin directions. So, $n_{\uparrow,m}n_{\downarrow,m} = 1$.
- ii. Expanding the second term first

$$(n_{i,\uparrow,1} + n_{i,\downarrow,1})(n_{i,\uparrow,2} + n_{i,\downarrow,2}) = n_{i,\uparrow,1}n_{i,\uparrow,2} + n_{i,\downarrow,1}n_{i,\uparrow,2} + n_{i,\uparrow,1}n_{i,\downarrow,2} + n_{i,\downarrow,1}n_{i,\downarrow,2}.$$

The expanded 4 terms are in different orbitals 1 and 2, so $(n_{i,\uparrow,1} + n_{i,\downarrow,1})(n_{i,\uparrow,2} + n_{i,\downarrow,2}) = 0$.

- iii. Obviously, $\vec{S}_1 \cdot \vec{S}_2 = 0$ since both electrons occupy the same orbital and one of $\vec{S}$ is zero.
- iv. For the last term, calculate the denominator for the second-order perturbation,

$$\Delta E = E_{\text{ext}} - E_{\text{int}}.$$

1. Initial state: There is one electron in the two neighbor site, then $E_{\text{int}} = (-\mu) + (-\mu) = -2\mu$.
2. Excited state: There are two electrons in one site, leaving another empty. Then a Coulomb potential is applied, $E_{\text{ext}} = \tilde{U} + (-\mu) + (-\mu) = U - 2\mu$.

This chemical potential term cancels in the energy denominator and therefore does not contribute.

In summary, the energy in this case is

$$E_{\text{same}} = U \cdot 1 + (U' - J_H/2) \cdot 0 - 2J_H \cdot 0 - 0 = U.$$

(b) **Electrons in different orbitals** Let the electron in orbital 1 has spin σ_1 , the electron in orbital 2 has spin σ_2 . The four terms are

- i. $n_{\uparrow,m}n_{\downarrow,m}$ is obviously zero, since they are at the same orbital.
- ii. In each bracket of $(n_{i,\uparrow,1} + n_{i,\downarrow,1})(n_{i,\uparrow,2} + n_{i,\downarrow,2})$, only one term can be 1 and another one should be 0. So, this term gives a value of $1 \times 1 = 1$.
- iii. The Hund term couples the spins of the two orbitals

$$\vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2}[(\vec{S}_1 + \vec{S}_2)^2 - \vec{S}_1^2 - \vec{S}_2^2] = \frac{1}{2}\left(|\vec{S}|^2 - \frac{3}{4} - \frac{3}{4}\right) = \frac{1}{2}\left(|\vec{S}|^2 - \frac{3}{2}\right),$$

where each $\vec{S}_m$ corresponds to spin- $\frac{1}{2}$ of the electron in orbital m , so $\vec{S}_m^2 = \frac{3}{4}$. The total spin $\vec{S} = \vec{S}_1 + \vec{S}_2$ has the

- i. Triplet: $|\vec{S}| = 1$, then $\vec{S}^2 = |\vec{S}|(|\vec{S}| + 1) = 2$, $\vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2}\left(2 - \frac{3}{2}\right) = \frac{1}{4}$.
- ii. Singlet: $|\vec{S}| = 0$, then $\vec{S}^2 = 0$, $\vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2}\left(0 - \frac{3}{2}\right) = -\frac{3}{4}$.
- iv. The same as the case (a). This term vanishes.

In summary, the energy in this case is

$$E_{\text{diff}} = \begin{cases} (U' - \frac{J_H}{2}) - 2J_H \times \frac{1}{4} = U' - J_H, & \text{Triplet,} \\ (U' - \frac{J_H}{2}) - 2J_H \times \left(-\frac{3}{4}\right) = U' + J_H, & \text{Singlet.} \end{cases}$$

Now, move on to the problems.

- I. Consider an electron hopping from site i to j (or inverse), one can have site j doubly occupied and leave site i empty (or inverse). The double occupancy have the energies

- (a) Same orbital: U . (b) Different orbitals, triplet: $U' - J_H$. (c) Different orbitals, singlet: $U' + J_H$.

which has been referred to in the preparation work: (a) - iv.

- i. Same orbital: $m_i = m_j = m$, with the energy denominator U .

When an electron from orbital m on site i hops to the same orbital of the site j , this gives double occupancy in the same orbital m on site j ; Also for the inverse. So, the perturbation can be written as

$$\mathcal{V} = \mathcal{V}_{i \rightarrow j} + \mathcal{V}_{j \rightarrow i}, \quad \mathcal{V}_{i \rightarrow j} = t_{ij} \sum_{\sigma} c_{j,\sigma,m}^{\dagger} c_{i,\sigma,m}, \quad \mathcal{V}_{j \rightarrow i} = t_{ij} \sum_{\sigma} c_{i,\sigma,m}^{\dagger} c_{j,\sigma,m}.$$

Due to the symmetry, the small hopping coefficients t_{ji} are the same in the two hopping directions. Since the orbital is fixed, due to Pauli exclusion, the initial state of an arbitrary single bond $\langle ij \rangle$ can be *only* the spin-singlet state

$$|S_{ij}\rangle = \frac{1}{\sqrt{2}}(|\uparrow_i \downarrow_j\rangle - |\downarrow_i \uparrow_j\rangle) = \frac{1}{\sqrt{2}}(c_{i,\uparrow}^{\dagger} c_{j,\downarrow}^{\dagger} - c_{i,\downarrow}^{\dagger} c_{j,\uparrow}^{\dagger})|0\rangle.$$

Applying $\mathcal{V}$ on it to obtain the numerator of the perturbed energy. In this case, consider each electron is moving in the same arbitrary orbit, then the orbital label can be omitted.

1. Hopping from i to j .

$$\mathcal{V}_{i \rightarrow j}|S_{ij}\rangle = \frac{t_{ij}}{\sqrt{2}} \sum_{\sigma} c_{j,\sigma}^{\dagger} c_{i,\sigma} (c_{i,\uparrow}^{\dagger} c_{j,\downarrow}^{\dagger} - c_{i,\downarrow}^{\dagger} c_{j,\uparrow}^{\dagger})|0\rangle.$$

The kernel of $\mathcal{V}_{i \rightarrow j}|S_{ij}\rangle$ is

$$\begin{aligned} \sum_{\sigma} [c_{j,\sigma}^{\dagger} c_{i,\sigma} c_{i,\uparrow}^{\dagger} c_{j,\downarrow}^{\dagger} - c_{j,\sigma}^{\dagger} c_{i,\sigma} c_{i,\downarrow}^{\dagger} c_{j,\uparrow}^{\dagger}] &= \sum_{\sigma} [c_{j,\sigma}^{\dagger} (\delta_{\sigma,\uparrow} - c_{i,\uparrow}^{\dagger} c_{i,\sigma}) c_{j,\downarrow}^{\dagger} - c_{j,\sigma}^{\dagger} (\delta_{\sigma,\downarrow} - c_{i,\downarrow}^{\dagger} c_{i,\sigma}) c_{j,\uparrow}^{\dagger}] \\ &= \sum_{\sigma} [\delta_{\sigma,\uparrow} c_{j,\sigma}^{\dagger} c_{j,\downarrow}^{\dagger} - \delta_{\sigma,\downarrow} c_{j,\sigma}^{\dagger} c_{j,\uparrow}^{\dagger}] = 2c_{j,\uparrow}^{\dagger} c_{j,\downarrow}^{\dagger}, \end{aligned}$$

where we used the anticommutative relation of Fermions

$$\{c_{i,\sigma}^{\dagger}, c_{j,\sigma'}^{\dagger}\} = 0, \quad \{c_{i,\sigma}, c_{j,\sigma'}^{\dagger}\} = \delta_{ij} \delta_{\sigma\sigma'}.$$

Hence, we can obtain

$$\mathcal{V}_{i \rightarrow j}|S_{ij}\rangle = \sqrt{2}t_{ij}|0_i; \uparrow_j, \downarrow_j\rangle = \sqrt{2}t_{ij}|\text{Doublon at } j, \text{ empty at } i\rangle.$$

According to the perturbation theory, in the numerator of the perturbed energy

$$\Delta E_a^{(2)} = - \sum_v \frac{|\langle v|\mathcal{V}|a\rangle|^2}{E_v - E_a}$$

contains all possible intermediate states that are not in the low-energy manifold. So, this case contributes to the numerator

$$|\langle \text{Doublon at } j, \text{ empty at } i | \mathcal{V}_{i \rightarrow j} | S_{ij} \rangle|^2 = 2t_{ij}^2.$$

The colored block indicates the action target of $\mathcal{V}_{i \rightarrow j}$.

2. Symmetrically, this case contributes

$$|\langle \text{Doublon at } i, \text{ empty at } j | \mathcal{V}_{j \rightarrow i} | S_{ij} \rangle|^2 = 2t_{ij}^2,$$

since they cause a symmetric result (See figures Fig. 1 and Fig. 3).

In summary, By substituting the contributions into the perturbation theory, the total energy correction of the case same orbital & singlet is

$$\Delta E_{\text{same, singlet}} = - \left(\frac{2t_{ij}^2}{U} + \frac{2t_{ij}^2}{U} \right) = - \frac{4t_{ij}^2}{U}. \quad (\text{A.1})$$

To obtain the perturbed Hamiltonian for this part, define the operators for filtering the states with the case of the same orbital

$$P_{ij}^{\text{same}} = \begin{cases} 1, & m_i = m_j, \\ 0, & \text{otherwise} \end{cases} = \frac{1}{2} + 2\tau_i^z \tau_j^z, \quad P_{ij}^{\text{diff}} = 1 - P_{ij}^{\text{same}} = \frac{1}{2} - 2\tau_i^z \tau_j^z.$$

where $\tau^z = \pm \frac{1}{2}$ is the orbital pseudospin, and operators for filtering the singlet and triplet states

$$P_{ij}^S = \frac{1}{4} - \vec{S}_i \cdot \vec{S}_j, \quad P_{ij}^T = \frac{3}{4} + \vec{S}_i \cdot \vec{S}_j$$

from the properties of the singlet $\vec{S}_i \cdot \vec{S}_j = -\frac{3}{4}$ and the triplet $\vec{S}_i \cdot \vec{S}_j = \frac{1}{4}$. Now, we can write the Hamiltonian for this part

$$\mathcal{H}_{\text{same}}^{ij} = - \frac{4t_{ij}^2}{U} P_{ij}^{\text{same}} P_{ij}^S.$$

ii. Different orbitals: $m_i \neq m_j$.

Consider an arbitrary initial state

1. Site i : Orbital $m = 1$, spin σ_1 ,

2. Site j : Orbital $m = 2$, spin σ_2 ,

which can be expressed as

$$|S_{ij}\rangle = c_{i,\sigma_1,1}^\dagger c_{j,\sigma_2,2}^\dagger |0\rangle,$$

and there are two allowed processes for it

1. Electron hops from i (orbital 1) to j (orbital 1).

2. Electron hops from j (orbital 2) to i (orbital 2).

They will also cause a symmetric result (see figures Fig. 2 and Fig. 4) like in part (i). So, analyze one process in parallel spins case and antiparallel spins case is enough.

1. Parallel Spins Case: Take the spin directions $\sigma_i = \sigma_j = \uparrow$.

The initial state now becomes

$$|\psi_{\text{para}}\rangle = c_{i,\uparrow,1}^\dagger c_{j,\uparrow,2}^\dagger |0\rangle.$$

In the first process

$$\begin{aligned} \mathcal{V}_{i \rightarrow j}^{(1)} |\psi_{\text{para}}\rangle &= t_{ij} \sum_{\sigma} c_{j,\sigma,1}^\dagger c_{i,\sigma,1} c_{i,\uparrow,1}^\dagger c_{j,\uparrow,2}^\dagger |0\rangle = t_{ij} \sum_{\sigma} c_{j,\sigma,1}^\dagger (\delta_{\sigma,\uparrow} - c_{i,\uparrow,1}^\dagger c_{i,\sigma,1}) c_{j,\uparrow,2}^\dagger |0\rangle \\ &= t_{ij} c_{j,\uparrow,1}^\dagger c_{j,\uparrow,2}^\dagger |0\rangle = t_{ij} |0_i; \uparrow_j \uparrow_j\rangle. \end{aligned}$$

This is a state with two electrons at j : One in orbital 1 parallels to another one in orbital 2, corresponding to a triplet configuration, and the second process will give a similar result. They will contribute to the perturbed energy

$$\Delta E_{\text{diff, para}} = -2 \times \frac{|\langle \uparrow_j \uparrow_j; 0_i | t_{ij} | 0_i; \uparrow_j \uparrow_j \rangle|^2}{E_{\text{diff, triplet}}} = -\frac{2t_{ij}^2}{U' - J_H}. \quad (\text{A.2})$$

2. Antiparallel Spins Case: Take the spin directions $\sigma_i = \uparrow, \sigma_j = \downarrow$.

The initial state now becomes

$$|\psi_{\text{anti}}\rangle = c_{i,\uparrow,1}^\dagger c_{j,\downarrow,2}^\dagger |0\rangle.$$

In the first process

$$\begin{aligned} \mathcal{V}_{i \rightarrow j}^{(1)} |\psi_{\text{anti}}\rangle &= t_{ij} \sum_{\sigma} c_{j,\sigma,1}^\dagger c_{i,\sigma,1} c_{i,\uparrow,1}^\dagger c_{j,\downarrow,2}^\dagger |0\rangle = t_{ij} \sum_{\sigma} c_{j,\sigma,1}^\dagger (\delta_{\sigma,\uparrow} - c_{i,\uparrow,1}^\dagger c_{i,\sigma,1}) c_{j,\downarrow,2}^\dagger |0\rangle \\ &= t_{ij} c_{j,\uparrow,1}^\dagger c_{j,\downarrow,2}^\dagger |0\rangle = t_{ij} |0_i; \uparrow_j \downarrow_j\rangle. \end{aligned}$$

This is a 50-50 mixture of triplet and singlet

$$|\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) + \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \right] = \frac{1}{\sqrt{2}} (|T\rangle + |S\rangle),$$

and the two parts will contribute

$$\Delta E_{\text{diff, anti}} = - \left(\frac{|\langle T | t_{ij} | T \rangle|^2}{U' - J_H} + \frac{|\langle S | t_{ij} | S \rangle|^2}{U' + J_H} \right) = -\frac{t_{ij}^2}{U' - J_H} - \frac{t_{ij}^2}{U' + J_H} \quad (\text{A.3})$$

to the correction.

In summary, to obtain the perturbed Hamiltonian for the case different orbitals, using the defined projectors and combining the two cases with the coefficients

$$\mathcal{H}_{\text{diff}}^{ij} = -\frac{2t_{ij}^2}{U' - J_H} P_{ij}^{\text{diff}} P_{ij}^T - \frac{2t_{ij}^2}{U' + J_H} P_{ij}^{\text{diff}} P_{ij}^S.$$

to make this part Hamiltonian satisfy at $P_{ij}^{\text{diff}} = 1$

$$\mathcal{H}_{\text{diff}}^{ij} = \begin{cases} -\frac{2t^2}{U' - J_H}, & P^T = 1, P^S = 0, \\ -\frac{t^2}{U' - J_H} - \frac{t^2}{U' + J_H}, & P^T = P^S = \frac{1}{2}. \end{cases}$$

By summing, we can obtain the compact form of the second-order perturbed Hamiltonian

$$\mathcal{H}_{ij}^{(2)} = -\frac{4t_{ij}^2}{U} P_{ij}^{\text{same}} P_{ij}^S - \frac{2t_{ij}^2}{U' - J_H} P_{ij}^{\text{diff}} P_{ij}^T - \frac{2t_{ij}^2}{U' + J_H} P_{ij}^{\text{diff}} P_{ij}^S.$$

Substituting the projection operators, we obtained the perturbed Hamiltonian in Kugel-Khomskii operator form

$$\begin{aligned} \mathcal{H}_{ij}^{(2)} = & -\frac{t_{ij}^2}{2} \left(\frac{1}{U} + \frac{2U' + J_H}{U'^2 - J_H^2} \right) + 2t_{ij}^2 \left(\frac{1}{U} - \frac{J_H}{U'^2 - J_H^2} \right) \vec{S}_i \cdot \vec{S}_j \\ & + 2t_{ij}^2 \left(-\frac{1}{U} + \frac{2U' + J_H}{U'^2 - J_H^2} \right) \tau_i^z \tau_j^z + 8t_{ij}^2 \left(\frac{1}{U} + \frac{J_H}{U'^2 - J_H^2} \right) (\vec{S}_i \cdot \vec{S}_j) (\tau_i^z \tau_j^z). \quad (\text{A.4}) \end{aligned}$$

- II. One can construct a Néel-type product state, but for realistic Hund's coupling J_H the ground state is typically a ferromagnetic spin order with staggered orbital order due to Hund-enhanced ferromagnetic exchange on inter-orbital bonds.

Proof. Evaluate the energy for the semi-classical ground energy of a bond $\langle ij \rangle$

$$E_{ij}^{\text{class}} = \langle \Psi_{\text{class}} | \mathcal{H}_{ij}^{(2)} | \Psi_{\text{class}} \rangle,$$

where the state can be written as

$$|\Psi_{\text{class}}\rangle = \bigotimes_i |\hat{n}_i\rangle_{\text{spin}} \otimes |\tau_i^z\rangle_{\text{orbital}}.$$

in which $\hat{n}_i$ denotes the spin direction, and τ_i^z denotes the orbital occupation: $\tau_i^z = \pm \frac{1}{2}$ for either orbital 1 or orbital 2. For a product state with fixed spin directions and orbital pseudospins

$$\langle \vec{S}_i \cdot \vec{S}_j \rangle = S^2 \cos \theta_{ij} = \frac{1}{4} \cos \theta_{ij}, \quad \langle \tau_i^z \tau_j^z \rangle = \tau_i \tau_j, \quad \tau_i^z = \pm \frac{1}{2},$$

where θ_{ij} denotes the angle between spin directions at site i and j . Thus, the expectation values of the projectors can be expressed as

$$\langle P_{ij}^S \rangle = \frac{1}{4} (1 - \cos \theta_{ij}), \quad \langle P_{ij}^T \rangle = \frac{1}{4} (3 + \cos \theta_{ij}), \quad \langle P_{ij}^{\text{same}} \rangle = \frac{1}{2} + 2\tau_i \tau_j, \quad \langle P_{ij}^{\text{diff}} \rangle = \frac{1}{2} - 2\tau_i \tau_j.$$

Now, the uncertain variables are θ_{ij} and $\tau_i \tau_j$. For simplicity, define the three exchange scales

$$J_1 \equiv \frac{4t_{ij}^2}{U}, \quad J_- \equiv \frac{2t_{ij}^2}{U' - J_H}, \quad J_+ \equiv \frac{2t_{ij}^2}{U' + J_H}.$$

Then, we can analyze bond energies in two different cases.

- (a) Same orbital bond: $\tau_i \tau_j = +\frac{1}{4}$. The projectors now become

$$\langle P_{ij}^{\text{same}} \rangle = \frac{1}{2} + 2 \times \frac{1}{4} = 1, \quad \langle P_{ij}^{\text{diff}} \rangle = \frac{1}{2} - 2 \times \frac{1}{4} = 0.$$

Then, the energy

$$E_{ij}^{\text{same}} = -\frac{4t_{ij}^2}{U} \langle P_{ij}^{\text{same}} \rangle \langle P_{ij}^S \rangle = -\frac{t_{ij}^2}{U} (1 - \cos \theta_{ij}) = -\frac{J_1}{4} (1 - \cos \theta_{ij}).$$

We can obtain the minimum (at $\theta_{ij} = \pi$) and the maximum (at $\theta_{ij} = 0$), between which the lattice will behave like an antiferromagnetic, driven by Hund's exclusion

$$E_{\min}^{\text{afm}} = -\frac{J_1}{2}, \quad E_{\max}^{\text{afm}} = 0. \quad \uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$$

(b) Different orbital bond: $\tau_i \tau_j = -\frac{1}{4}$. The projectors

$$\langle P_{ij}^{\text{same}} \rangle = \frac{1}{2} - 2 \times \frac{1}{4} = 0, \quad \langle P_{ij}^{\text{diff}} \rangle = \frac{1}{2} + 2 \times \frac{1}{4} = 1.$$

Then, the energy

$$\begin{aligned} E_{ij}^{\text{diff}} &= -\frac{2t_{ij}^2}{U' - J_H} \langle P_{ij}^{\text{diff}} \rangle \langle P_{ij}^T \rangle - \frac{2t_{ij}^2}{U' + J_H} \langle P_{ij}^{\text{diff}} \rangle \langle P_{ij}^S \rangle \\ &= -\frac{t_{ij}^2}{2} \left[\frac{3 + \cos \theta_{ij}}{U' - J_H} + \frac{1 - \cos \theta_{ij}}{U' + J_H} \right] = -\frac{3J_- + J_+}{4} - \frac{J_- - J_+}{4} \cos \theta_{ij}. \end{aligned}$$

Since $U' - J_H < U' + J_H$, so $J_- > J_+$, the coefficient of $-\cos \theta_{ij}$ is positive. Then, we can obtain the minimum (at $\theta_{ij} = 0$) and the maximum (at $\theta_{ij} = \pi$), between which the lattice will behave like ferromagnetic

$$E_{\min}^{\text{fm}} = -J_-, \quad E_{\max}^{\text{fm}} = -\frac{1}{2}(J_- + J_+). \quad \uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$$

To judge which case has lower energy, just compare their minimal energies

$$E_{\min}^{\text{afm}} = -\frac{J_1}{2} = -\frac{2t_{ij}^2}{U}, \quad E_{\min}^{\text{fm}} = -J_- = -\frac{2t_{ij}^2}{U' - J_H},$$

the lower one will win. Due to the rotational invariance in orbital space, $U' = U - 2J_H$, even without this relation, since the positive Hund's coupling, the ferromagnetic state has the lowest energy, which will be denoted as the ground state.

Hence, a Néel-type product state can be constructed, but it is not the ground state for physically realistic parameters with nonzero Hund's coupling. Except *in the absence of Hund's coupling* ($J_H = 0$), *the two energies become equal, and the Néel state would be both the ground state and also degenerate with the other state.* $\square$

III. (a) The retarded Green's function for orbital m , spin σ on site i is

$$G_{m,\sigma}^i(\omega) = \langle \langle c_{i,m,\sigma}; c_{i,m,\sigma}^\dagger \rangle \rangle_\omega.$$

In the atomic limit ($U \gg t$), one can compute the Green's function using Lehmann representation

$$G_{m,\sigma}^i(\omega) = \sum_n \frac{|\langle n | c_{i,m,\sigma}^\dagger | i \rangle|^2}{\omega - (E_n - E_0) + i0^+} + \sum_n \frac{|\langle n | c_{i,m,\sigma} | i \rangle|^2}{\omega + (E_n - E_0) + i0^+},$$

where $|i\rangle = c_{i,m_0,\sigma_0}^\dagger |0\rangle$ is the initial state for site i : m_0 can be taken as 1 or 2, and σ_0 can be taken as $\uparrow$ or $\downarrow$. Consider applying an annihilation or a creation operator for removal or addition to a site, due to the difference between m and m_0 , σ and σ_0 , it can be divided into 4 cases

i. Same orbital, same spin ($m = m_0, \sigma = \sigma_0$).

1. Removal: $c_{i,m_0,\sigma_0} c_{i,m_0,\sigma_0}^\dagger |0\rangle = |0\rangle$.

2. Addition: Blocked by the Pauli exclusion principle, no addition pole.

The matrix element is 1, the pole is $\omega = 0$. The Green's function is

$$G^{\text{same, same}} = \frac{1}{\omega + i0^+}.$$

ii. Same orbital, different spin ($m = m_0, \sigma = -\sigma_0$).

1. Removal: $c_{i,m_0,-\sigma_0} c_{i,m_0,\sigma_0}^\dagger |0\rangle = 0$ due to there is no spin $-\sigma_0$ at m_0 orbital for removal.

2. Addition: $c_{i,m_0,-\sigma_0}^\dagger c_{i,m_0,\sigma_0}^\dagger |0\rangle = |m_0, \sigma_0; m_0, -\sigma_0\rangle$.

The matrix element is 1, the pole is $\omega = U$. The Green's function is

$$G^{\text{same, diff}} = \frac{1}{\omega - U + i0^+}.$$

iii. Different orbital, same spin ($m \neq m_0, \sigma = \sigma_0$).

1. Removal: $c_{i,m,\sigma_0} c_{i,m_0,\sigma_0}^\dagger |0\rangle = 0$ due to there is no spin at m orbital for removal.

2. Addition: $c_{i,m,\sigma_0}^\dagger c_{i,m_0,\sigma_0}^\dagger |0\rangle = |m_0, \sigma_0; m, \sigma_0\rangle$, triplet.

The matrix element is 1, the pole is $\omega = U' - J_H$. The Green's function is

$$G^{\text{diff, same}} = \frac{1}{\omega - (U' - J_H) + i0^+}.$$

iv. Different orbital, different spin ($m \neq m_0, \sigma = -\sigma_0$).

1. Removal: $c_{i,m,-\sigma_0} c_{i,m_0,\sigma_0}^\dagger |0\rangle = 0$ due to there no spin at m orbital for removal.

2. Addition: $c_{i,m,-\sigma_0}^\dagger c_{i,m_0,\sigma_0}^\dagger |0\rangle = |m_0, \sigma_0; m, -\sigma_0\rangle = \frac{1}{\sqrt{2}}(|T\rangle + |S\rangle)$, 50-50 mixture of triplet and singlet.

The matrix element for both triplet and singlet is $1/2$, the poles are $\omega = U' - J_H$ and $\omega = U' + J_H$. The Green's function is

$$G^{\text{diff,diff}} = \frac{1}{2} \frac{1}{\omega - (U' - J_H) + i0^+} + \frac{1}{2} \frac{1}{\omega - (U' + J_H) + i0^+}.$$

(b) Starting from the effective Hamiltonian in (A.4), its expectation can be written as

$$\langle \mathcal{H}_{ij}^{(2)} \rangle = J_s m^2 + J_\tau p^2 + J_{S_\tau} m^2 p^2 + \text{Constant},$$

where

$$\begin{aligned} J_s &= 2t_{ij}^2 \left(\frac{1}{U} - \frac{J_H}{U'^2 - J_H^2} \right), & J_\tau &= 2t_{ij}^2 \left(-\frac{1}{U} + \frac{2U' + J_H}{U'^2 - J_H^2} \right), & J_{S_\tau} &= 8t_{ij}^2 \left(\frac{1}{U} + \frac{J_H}{U'^2 - J_H^2} \right), \\ m^2 &= \langle \vec{S}_i \cdot \vec{S}_j \rangle, & p^2 &= \langle \tau_i^z \tau_j^z \rangle, & \text{Constant} &= -\frac{t_{ij}^2}{2} \left(\frac{1}{U} + \frac{2U' + J_H}{U'^2 - J_H^2} \right). \end{aligned}$$

Summing over bonds gives the mean energy per site

$$U_{\text{Mean Field}} = \frac{z}{2}(J_S m^2 + J_\tau p^2 + J_{S_\tau} m^2 p^2),$$

where z is the coordination number, and a factor $\frac{1}{2}$ is applied to avoid doubling when summing over $\langle ij \rangle$. Then, we can obtain the mean-field free energy per site by definition

$$F_{\text{Mean field}} = U_{\text{Mean Field}} - T s(m, p) = \frac{z}{2}(J_S m^2 + J_\tau p^2 + J_{S_\tau} m^2 p^2) - T[s(m) + s(p)].$$

Then calculate the entropy. Denote P_+ , P_- the probability of $S^z = +\frac{1}{2}$ and $S^z = -\frac{1}{2}$ respectively. Then, the expectation of the magnetization is

$$m = \langle S^z \rangle = \left(\frac{1}{2}\right)P_+ + \left(-\frac{1}{2}\right)P_-.$$

Due to the normalization $P_+ + P_- = 1$, we can obtain

$$P_+ = \frac{1}{2} + m, \quad P_- = \frac{1}{2} - m.$$

Then, substitute them into the definition of the entropy

$$\begin{aligned} s_S(m) &= -\sum_i P_i \ln P_i = -[P_+ \ln P_+ + P_- \ln P_-] \\ &= -\left[\left(\frac{1}{2} + m\right) \ln\left(\frac{1}{2} + m\right) + \left(\frac{1}{2} - m\right) \ln\left(\frac{1}{2} - m\right)\right]. \end{aligned}$$

which satisfies $s_S(\pm\frac{1}{2}) = 0$ for fully polarized states. For minimization, calculating the derivative of the entropy for magnetization, and similarly for polarization

$$\frac{\partial s_S}{\partial m} = -\ln\left(\frac{1+2m}{1-2m}\right) = -2 \operatorname{th}^{-1}(2m), \quad \frac{\partial s_S}{\partial p} = -\ln\left(\frac{1+2p}{1-2p}\right) = -2 \operatorname{th}^{-1}(2p).$$

Then, minimize the free energy with respect to m and p : Let $\partial F_{\text{Mean field}}/\partial m = 0$ first

$$z(J_S m + J_{S_\tau} m p^2) - T \frac{\partial s_S}{\partial m} = 0.$$

Substituting $\partial s_S/\partial p$, one can obtain

$$z(J_S + J_{S_\tau} p^2)m = 2T \operatorname{th}^{-1}(2m),$$

from which can solve the equation for magnetization

$$m = \frac{1}{2} \operatorname{th} \left[\frac{z}{2T} (J_S + J_{S_\tau} p^2 m) \right].$$

Since m and p are equivalent, yields

$$p = \frac{1}{2} \operatorname{th} \left[\frac{z}{2T} (J_S + J_{S_\tau} m^2 p) \right].$$

(c) See if my achievements are qualified for writing a paper :-)

Problem 0.2. Single band Hubbard model at HALF-filling with resonant valence bond correlations. (ongoing work) In the lecture, we discussed how to solve for single particle Green's functions as perturbation to a Néel product state. However, a more legitimate calculation should be a degenerate perturbative calculation, realized by starting with a parameterized superposition of the two-fold degenerate Néel states. The superposition construction naturally introduces the z -component of resonant valence bond, but not the transverse (x & y) components.

Objectives:

- (a) **Minimal I.** Write down the parameterized superposition of the two-fold degenerate Néel states on square lattice; compute the unperturbed static correlation functions, including single-particle (note now there is nonlocal correlations), spin-spin and vertex-like ones, as functions of the parameters.
 - (b) **Minimal II.** Obtain the “atomic” limit solutions for single particle GFs; compare them with the Néel state case.
 - (c) **Advanced.**
 - i. Perform the second-order perturbation calculation for energy (i.e. for the hopping term), you'd recover the superexchange magnetic energy in its SU(2) form (note in Néel case, we only have the Ising form, since the atomic state is always fully polarized); discuss if there is any correction of the same order to the U-term? (Hint: You need to take a third order time-derivative on GFs).
 - ii. Try to derive the single particle GFs at involving the RVB correlations; plot the spectral functions; check with *K. Kurokawa et al., “Unveiling phase diagram of the lightly doped high- T_c cuprate superconductors with disorder removed”, Nature communications 14, 4064 (2023).*
 - iii. Although we are working on solutions for the insulating phase, recent experimental evidence suggests that, a complete clean system at finite doping may be just a simple “RVB” metal with Fermi-liquid-like Fermi pockets *K. Kurokawa et al., “Unveiling phase diagram of the lightly doped high- T_c cuprate superconductors with disorder removed”, Nature communications 14, 4064 (2023).* If you have solve #2, try just shift the chemical potential right across the lower Hubbard band and focusing on this branch of dispersion (Hubbard band).
 - i. Write down its self-energy, and discuss its quasiparticle life-time, is it Fermi-liquid-like?
 - ii. Try to compute its superconductivity susceptibility:
 - A. The bare particle-particle bubble;
 - B. The RPA form.
- Do find superconductivity instability, i.e. divergence in the RPA form solution? If you reach this point, come and talk to for our Nobel prize winning paper.

References

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